

RAHUL RAVI VK

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WORK EXPERIENCE

Research Intern | *Telehealth and Robotics Lab, Children's National Medical Center, Washington, D.C.* May 2026 – Aug 2026

- Working under Dr. Reza Monfaredi (Sheikh Zayed Institute for Pediatric Surgical Innovation) on medical robotics and smart systems research.
- Contributing to a smart wheelchair customized for children with Cerebral Palsy, a surgical tool-sorting robot using UR5, and a telerobotic system using the UR3 robot with Geomagic Touch haptic device.

Assistant Manager, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* May 2023 – Jul 2025

- Led automation and predictive maintenance across **Honeywell Experion PKS C300 DCS** and **Siemens PLC** arrays; deployed pump torque-converter speed control and PLC bistable-relay emulation, reducing equipment downtime **50%** and false trips **90%**.
- Directed cross-functional cutovers (Mechanical, Electrical, Production, Civil) and planned annual shutdown/turnaround from scheduling through production restart.

Senior Engineer, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* Oct 2019 – Apr 2023

- Owned DCS and safety-system operations supporting **4000+ control loops**; managed **Honeywell Safety Manager (ESD)** and Experion DCS at **99.5%+** availability with RAID5 storage administration and one-to-one spare-station redundancy.
- Executed a 5-hour online tuning session on a safety-critical flow loop, eliminating oscillation and overshoot without interrupting production; reduced process downtime **30%** via PID retuning and actuator nonlinearity analysis.

Graduate Engineer Trainee, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* Oct 2018 – Oct 2019

- Built field-instrumentation foundation across live process lines: installed, calibrated, and validated 20+ transmitter types to $\pm 0.075\%$ FS across 4–20 mA / HART loops.
- Owned full automation lifecycle: field wiring, calibration, PLC sequencing, safety interlocks, alarm rationalization, loop checking, FAT/SAT, and predictive maintenance.

AUTONOMY & ROBOTICS PROJECTS

Autonomous Block Retrieval Robot *OpenCV, Raspberry Pi, Arduino, PID*

- Built a fully autonomous robot from bare hardware to navigate a 10×10 ft arena, detect blocks in Red→Green→Blue sequence, pick and deliver **9/9 blocks in 7:21**; implemented HSV/contour tracking, bounding-box distance estimation, PID centering, visual servoing, servo gripper pickup confirmation, IMU–encoder fusion, and waypoint-based obstacle avoidance.

TurtleBot4 Real-Time Perception Pipeline *ROS 2, OpenCV, TurtleBot4, Classical CV*

- Developed a ROS 2 vision-control pipeline with four real-time perception tasks: arrow path following, UMD logo stop (SIFT+RANSAC), dynamic obstacle TTC stop (Farneback optical flow), and horizon overlay.
- Benchmarked ResNet18 and YOLO offline; deliberately chose classical CV after evaluating deployment latency on-robot—a direct example of evaluating AI solutions against real-world constraints.

Multi-Robot Path Planning — Anytime RRT* + ORCA *Python, Webots, LiDAR, TurtleBot3*

- Implemented a hierarchical planner for two TurtleBot3 robots: Anytime RRT* global planning with SciPy cKDTree nearest-neighbor search, 15% goal-biased sampling, in-flight rewiring; ORCA local velocity correction with pairwise collision avoidance and continuous LiDAR obstacle updates. Achieved **100% mission success** and **zero inter-robot collisions** across 10 paired runs vs. 4 collisions for plain RRT.

Optimal Control for Nonholonomic Mobile Robot *MATLAB, Genetic Algorithms, Adaptive PID*

- Reproduced Park et al. IEEE Access (2024): GA-optimized kinematic gains across 72 postures, 2D cubic-spline gain scheduling, and gradient-descent adaptive PID with Lyapunov stability verification; matched **12–16 s** reported arrival times.

EDUCATION

M.Eng. in Robotics | University of Maryland, College Park Sep 2025 – May 2027

GPA: 3.9/4.0 | Coursework: Perception for Autonomous Robots, Planning for Autonomous Robots, Control of Robotic Systems, Autonomous Robots, Robot Modeling, Industrial AI

B.Eng. in Instrumentation & Control | L.D. College of Engineering, Gujarat, India Aug 2014 – May 2018

GPA: 3.3/4.0 | GATE All India Rank 96

TECHNICAL SKILLS

- Autonomy & Robotics Frameworks:** ROS 2 Humble, MoveIt 2, Gazebo, ros2_control, URDF/Xacro, RViz2, Nav2, tf2, Docker, Linux/Ubuntu
- Perception & ML:** OpenCV, PyTorch, RGB/HSV segmentation, image classification (CNNs), feature matching, camera calibration, homography, model latency benchmarking
- Planning & Navigation:** A*, RRT*, Dijkstra, PRM, potential fields, pure pursuit, path smoothing, sim-to-real transfer
- Controls & Estimation:** PID, cascade control, LQR, LQG, MPC, Luenberger observers, Kalman filtering, state-space modeling, gain scheduling
- Embedded & Hardware:** BNO055 IMU, wheel encoders, ultrasonic sensors, GPIO, UART/I2C/serial, PWM motor control, differential drive, sensor fusion
- Industrial Automation:** Honeywell Experion PKS C300 DCS, Siemens PLC, SCADA, Safety Manager (ESD), 4–20 mA / HART, FAT/SAT, safety interlocks, TwinCat 3, Codesys
- Programming & Tools:** Python, C++17, MATLAB/Simulink, NumPy, Pandas, Bash, WebSockets, Git, IKPy, SolidWorks, Blender

CERTIFICATIONS

Siemens – Basics of PLC & SCADA | Honeywell – Experion PKS C300 Fundamentals